Generative modeling for speaker diarization

Second year CSI presentation

Sicheng Mao

2025/10/1







INSTITUT POLYTECHNIQUE DE PARIS

Outline

1. Overview
2. First Work
    1. Extend PLDA to non-linear analog
    2. Training
    3. Scoring on hidden space
    4. Experiment and Results
3. Second Work
    1. Joint separation and diarization
    2. Heavy-tailed modeling for source power specturm density
    3. Training
    4. Experiment and Results
4. Trainings and Teachings
5. Plans

Overview

This year we focus on two works:

- Regularize deep speaker embedding by hierarchical variational autoencoder
- Neural multichannel distant speaker diarization with heavy-tailed source separation model

The first work was submitted to WASPAA2025 and later APSIPA2025, but both of the conferences have rejected it due to the unsatisfying results. The second work have been just submitted to ICASSP2026.

First Work

Issue

There are two common scoring backends in speaker verification task,

- cosine similarity: $\frac{x_1^T x_2}{||x_1||||x_2||}$
- probabilistic linear discriminant analysis: suppose the following speaker embedding generation model and compute the similarity by hypothesis test based on likelihood.

$$P(\mathbf{y}) = \mathcal{N}(\mathbf{y} \mid \mathbf{m}, \mathbf{\Phi}_b)$$
$$P(\mathbf{x} \mid \mathbf{y}) = \mathcal{N}(\mathbf{x} \mid \mathbf{y}, \mathbf{\Phi}_w)$$

However, for deep speaker embeddings, PLDA is no more performing as the simplest cosine similarity. We guess it’ s because the deep speaker embedding is extracted from a neural network which follows badly the Gaussianity assumption of PLDA.

Extend PLDA to non-linear analog

Therefore, we consider to extend the PLDA to a non-linear analog to better cope with the deep speaker embedding distribution.

Formulation

Instead of Probablistic Linear discriminant analysis, we replace the linear dependency of utterance to speaker by non-linear one, i.e.

$$\mathbf{y}_i \sim p(\mathbf{y}) = \mathcal{N}(\mathbf{y}; \mathbf{0}, \mathbf{I}), \tag{1}$$

$$\mathbf{z}_{ij}|\mathbf{y}_i = \mathbf{y} \sim p(\mathbf{z}|\mathbf{y}) = \mathcal{N}(\mathbf{z}; f_{\boldsymbol{\theta}}(\mathbf{y}), \mathbf{I}), \tag{2}$$

$$\mathbf{x}_{ij} = g_{\boldsymbol{\theta}}(\mathbf{z}_{ij}), \tag{3}$$

where f_θ and g_θ is generally non-linear and we keep the covariances independent of speakers and normalized.

Training

The ELBO function for the observed N_i utterances $\mathbf{X}_i = \{\mathbf{x}_{ij}\}_{j=1}^{N_i}$ with hidden utterances $\mathbf{Z}_i = \{\mathbf{z}_{ij}\}_{j=1}^{N_i}$ from speaker $\mathbf{y}_i$ of the model:

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{X}) := \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{y}, \mathbf{Z}|\mathbf{X})}[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{X}, \mathbf{Z}, \mathbf{y})}{q_{\boldsymbol{\phi}}(\mathbf{y}, \mathbf{Z}|\mathbf{X})}], \tag{4}$$

$$= \mathbb{E}_{q(\mathbf{y}, \mathbf{Z}|\mathbf{X})}\left[\log p(\mathbf{X}|\mathbf{Z})\right] - \mathrm{KL}(q(\mathbf{y}, \mathbf{Z}|\mathbf{X})||p(\mathbf{y}, \mathbf{Z})), \tag{5}$$

By top-down inference: $q(\mathbf{y}, \mathbf{Z}|\mathbf{X}) = q(\mathbf{y}|\mathbf{X})q(\mathbf{Z}|\mathbf{y}, \mathbf{X})$

$$\begin{aligned} &\mathrm{KL}(q(\mathbf{y}, \mathbf{Z}|\mathbf{X})||p(\mathbf{y}, \mathbf{Z})) = \\ &\mathrm{KL}(q(\mathbf{y}|\mathbf{X})||p(\mathbf{y})) + \mathbb{E}_{q(\mathbf{y}|\mathbf{X})}[\mathrm{KL}(q(\mathbf{Z}|\mathbf{y}, \mathbf{X})||p(\mathbf{Z}|\mathbf{y}))]. \end{aligned} \tag{6}$$

Factorize the speaker conditional KL by the indepency between utterances from the same speaker.

$$\mathrm{KL}(q(\mathbf{Z}|\mathbf{y},\mathbf{X})||p(\mathbf{Z}|\mathbf{y})) = \sum_{j=1}^{N_i} \mathrm{KL}(q(\mathbf{z}_j|\mathbf{y},\mathbf{X})||p(\mathbf{z}_j|\mathbf{y})) \tag{7}$$

The ELBO for the whole dataset of all speakers is the sum of individual ELBOs, by the independency between speakers:

$$\log p_{\boldsymbol{\theta}}(\mathbf{X}) = \sum_{i=1}^{S} \log p_{\boldsymbol{\theta}}(\mathbf{X}_i) \geq \mathcal{L}(\boldsymbol{\theta},\boldsymbol{\phi};\mathbf{X}) = \sum_{i=1}^{S} \mathcal{L}(\boldsymbol{\theta},\boldsymbol{\phi};\mathbf{X}_i) \tag{8}$$

Scoring on hidden space

Once we get our regularized embeddings y, we can perform hypothesis test as follows:

$$
\begin{aligned}
R(\mathbf{x}_1, \mathbf{x}_2) &= \frac{P\left(\mathbf{x}_1, \mathbf{x}_2 | \mathcal{T}, \theta\right)}{P\left(\mathbf{x}_1, \mathbf{x}_2 | \mathcal{N}, \theta\right)} = \frac{\int p(x_1|y)p(x_2|y)p(y)dy}{p(x_1)p(x_2)} \\
&= \int \frac{p(x_1, y)p(x_2, y)}{p(x_1)p(x_2)p(y)}dy = \int \frac{p(y|x_1)p(y|x_2)}{p(y)}dy \\
&\approx \int \frac{q(y|x_1)q(y|x_2)}{q(y)}dy = x_1^T x_2 + const.
\end{aligned}
$$

Such simplification of score is due to our assumptions on the identity covariances. If we also perform the common length-normalization trick, we conclude that the scoring of our regularized embedding reduced to the cosine similarity on hidden space Y.

Experiment and Results

We perform regularizations over four kinds of the embeddings on two different dataset, however the results are not promising compared to the baselines, possibly due to KL-vanishing problem in HVAE training.

Table 1: Equal error rate (EER%) on different datasets with Sidiar extractor

Dataset	Acoustic env.	o-o	o-c	o-m	o-t	c-c	c-m	c-t	m-m	m-t	t-t
Voxceleb1o	x-cos	**2.14**	3.96	4.73	9.95	5.20	5.85	10.84	6.33	11.27	14.61
	x-plda	2.52	**3.74**	**4.35**	**7.88**	**4.98**	**5.47**	**9.01**	**5.85**	**9.50**	**11.88**
	y-cos	3.16	5.34	6.31	11.61	6.55	7.23	12.51	7.84	12.89	15.61
CNCeleb	x-cos	19.43	23.38	24.69	29.21	24.45	25.62	30.33	25.64	30.38	31.98
	x-plda	18.96	**21.73**	**22.37**	**25.13**	**22.16**	**22.83**	**25.30**	**22.48**	**25.22**	**25.10**
	y-cos	**18.60**	22.69	23.47	28.06	23.39	24.50	29.24	24.42	29.37	29.97

'o','c','m','t' stands for the original data, car reverberation, meeting room reverberation and theater reverberation respectively. For example, 'o-c' means the enrollment audio is original and the test audio is in car reverberation. The best score is in bold where as the second best score is underlined.

A possible improvement is to use a better training strategy (e.g. diffusion or flow matching) to solve the hierarchical Gaussian model. There is a similar work on Interspeech2025 called SEED.

Second Work

The biggest challenges in speaker diarization task are:

- complex acoustic environment
- overlap speeches

Intuitively, if we can well separate a speech mixture, it would reduce the speaker diarization to a simpler voice activity detection task. Therefore we would like to learn both separation and diarization jointly.

A previous attempt: the neural FCASA

Joint separation and diarization

Generative model

Source model by variance modeling

$$s_{nft} \sim \mathcal{N}_{\mathbb{C}}(0, \lambda_{nft})$$

Mixture model with speaker activity temporal mask

$$\mathbf{x}_{ft} = \sum_{n=1}^{N} u_{nt} \mathbf{x}_{nft} = \sum_{n=1}^{N} u_{nt} \mathbf{a}_{nf} s_{nft}$$

Joint diagonalizable spatial covariance matrix for computational efficiency

$$\mathbf{a}_{nf} \mathbf{a}_{nf}^{\mathsf{H}} = \mathbf{H}_{nf} = \mathbf{Q}_f^{-1} \operatorname{diag}(\mathbf{w}_{nf}) \mathbf{Q}_f^{-\mathsf{H}}$$

Heavy-tailed modeling for source power specturm density

We replace the source model by heavy-tailed models for better robustness.

A heavy-tailed extension can be simply formulated by adding a random positive impulse variable r_{ft} to the Gaussian variance, leading to a (circularly-symmetric) Gaussian Scalar Mixture~(GSM) model:

$$s_{nft}|r_{ft} \sim \mathcal{N}(0, r_{ft}\lambda_{nft})$$

where we can assume different prior distribution for r_{ft}. By marginalizing out it, we can get a specific heavy-tailed model for the mixture signal.

Particularly, we are interested in

$$\begin{aligned}
\mathbf{x}_{ft} &\sim \mathcal{GG}^{\beta}_{\mathbb{C}}(0, \mathbf{Y}_{ft}) \quad (\text{Leptokurtic}), \\
\mathbf{x}_{ft} &\sim \mathcal{T}^{\nu}_{\mathbb{C}}(0, \mathbf{Y}_{ft}) \qquad (\text{Student-t})
\end{aligned}$$

where $\beta \in (0, 2], \nu \in \mathbb{R}^+$ are shape parameters for Leptokurtic and Student-t distribution respectively. Note that the GSM reduces to a Gaussian model hence to the vanilla neural FCASA when r_{ft} is

Training

We are going to solve the model by maximizing its ELBO function

$$\log p_\theta(\mathbf{X}) = \log \int p_\theta(\mathbf{X}, \mathbf{U}, \mathbf{Z}) d\mathbf{U} d\mathbf{Z} \geq \underbrace{\mathbb{E}_{q_\phi(\mathbf{U}|\mathbf{X}) q_\phi(\mathbf{Z}|\mathbf{X})}(\log p_\theta(\mathbf{X} \mid \mathbf{U}, \mathbf{Z}))}_{\triangleq \mathcal{L}_1}$$

$$\underbrace{-\mathrm{KL}[q_\phi(\mathbf{Z} \mid \mathbf{X}) \mid\mid p_\theta(\mathbf{Z})]}_{\triangleq \mathcal{L}_2} \underbrace{-\mathrm{KL}[q_\phi(\mathbf{U} \mid \mathbf{X}) \mid\mid p_\theta(\mathbf{U})]}_{\triangleq \mathcal{L}_3}$$

where

$$\mathcal{L}_1 \overset{Leptokurtic}{=} \sum_{f,t=1}^{F,T} \left[-\left(\sum_{m=1}^{M} \frac{|\tilde{x}_{ftm}|^2}{\tilde{y}_{:ftm}} \right)^{\frac{\beta}{2}} - \sum_{m=1}^{M} \log \tilde{y}_{:ftm} \right] + T \sum_{f=1}^{F} \log |\mathbf{Q}_f \mathbf{Q}_f^{\mathsf{H}}|$$

$$\overset{Student-t}{=} \sum^{F,T} \left[-\left(\frac{\nu}{2} + M \right) \log \left(1 + \frac{2}{\nu} \sum^{M} \frac{|\tilde{x}_{ftm}|^2}{\tilde{y}_{:ftm}} \right) - \sum^{M} \log \tilde{y}_{:ftm}, \right] + T \sum^{F} \log |\mathbf{Q}_f \mathbf{Q}_f^{\mathsf{H}}|$$

$\mathcal{L}_2$ is KL divergence between two Gaussians and hence can be computed in closed form.

As to the third term, since we don't assume a prior model for speaker activity $\mathbf{U}$, we directly minimize the KL-divergence between the empirical posterior distribution and the surrogate, which is equivalent to minimizing the binary cross entropy:

$$\mathcal{L}_3 \approx -\sum_{n,t=1}^{N,T} \mathrm{BCE}[u_{nt}|\eta_{\phi,nt}].$$

Finally, we combine the three terms with scaling hyperparameters.

$$\mathcal{L} = \frac{1}{TF}\left(\mathcal{L}_1 + \delta\mathcal{L}_2\right) + \gamma\frac{1}{TN}\mathcal{L}_3,$$

Experiment and Results

Model	Shape param.	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
Gaussian (baseline)	-	Forgiving	5.59	8.16	0.73	14.48	13.50
		Fair	8.61	6.33	0.79	15.73	23.16
		Full	10.77	6.81	1.16	18.73	27.65
		Overlap	19.40	3.80	0.90	24.11	26.53
Leptokurtic	0.4	Forgiving	7.98	6.23	0.86	15.07	15.81
		Fair	11.52	4.97	0.92	17.41	27.33
		Full	13.85	5.42	1.28	20.54	31.52
		Overlap	24.07	3.05	0.90	28.02	31.35
	0.8	Forgiving	6.14	7.84	0.83	14.81	13.98
		Fair	9.35	6.29	0.96	16.61	24.36
		Full	11.63	6.82	1.30	19.75	28.70
		Overlap	20.79	4.20	0.96	25.95	28.10
	1.0	Forgiving	5.45	7.36	0.79	13.60	13.01
		Fair	8.84	5.76	0.80	15.41	22.66
		Full	11.08	6.26	1.13	18.47	27.38
		Overlap	20.13	3.52	0.80	24.45	27.02
	1.2	Forgiving	5.13	9.12	0.86	15.11	13.90
		Fair	8.30	6.80	0.95	16.06	23.59
		Full	10.38	7.58	1.34	19.30	28.17
		Overlap	18.98	4.11	1.03	24.12	26.78
	1.6	Forgiving	6.04	6.65	0.71	13.39	13.32
		Fair	9.55	5.08	0.75	15.39	23.33
		Full	11.82	5.65	1.06	18.52	27.92
		Overlap	21.09	3.10	0.74	24.93	27.82
Student-t	0.1	Forgiving	5.04	**5.59**	**0.61**	**11.24**	**10.78**
		Fair	7.49	**4.42**	**0.62**	**12.53**	**18.38**
		Full	9.23	5.86	**0.92**	**16.01**	**23.45**
		Overlap	16.18	**3.18**	**0.67**	20.03	21.99
	1	Forgiving	**4.30**	6.44	0.67	11.41	11.04
		Fair	**6.92**	5.02	0.69	12.62	18.80
		Full	**8.55**	6.52	0.98	16.05	23.82
		Overlap	**15.57**	3.44	0.70	**19.71**	**21.92**
	10	Forgiving	5.45	7.58	0.78	13.80	12.17
		Fair	8.55	6.17	0.80	15.52	20.98
		Full	10.93	5.99	1.07	17.99	25.95
		Overlap	19.76	3.71	0.74	24.21	26.01
	100	Forgiving	5.57	6.78	0.72	13.08	12.48
		Fair	8.93	5.39	0.80	15.12	22.09
		Full	11.25	5.96	1.09	18.30	26.82
		Overlap	20.12	3.55	0.78	24.44	26.72

We trained 9 heavy-tailed models and evaluate it on the AMI corpus with DER and JER score under 4 different setups. We improved the baseline by a large margin particularly for Student-t distribution with $\beta = 0.1, 1$

Future work: generalize for more heavy-tailed models and introduce a prior model for speaker activity

Trainings and Teachings

The courses I took by the end of the second year amounts to 133 hours (63 hours of scientific course, 70 hours of transversal skills): the following course was what I took in the second year

- Deep Learning Optimisé sur Jean Zay, 32 h (scientific)

which have already fulfilled the training requirements.

During my second year of PhD, I also taught some pratical session of the following courses (all in Telecom Paris) as well as supervised some project/internship:

- CSC_5AI06_TP - Deep learning I(12h)
- APM_4AI01_TP - Représentations des signaux(3h)
- APM_0EL07_TP - Logic, Knowledge Representation and Probabilities(10.5h)
- Master 1 Data AI Research project (Cosupervised Taiwei Wu with Mathieu)
- Engineer 1A student internship (Cosupervised

Plans

For Interspeech 2026 or next year conferences:

- Generalize the second work by more heavy-tailed models and introduce a prior model for speaker activity label.
- Try the diffusion or the flow matching for the hierarchical Gaussian model training or try the hyperspherical distribution as a training target.

For journal paper in next year:

- If everything works well, extend the second work to a journal paper after the conferences end

For the defense:

- Write the manuscript in the last 3-4 months.

END